

DevOps Lab-3

Code a Dockerized python flask or node.js application.

Program Structure.

- |
- |— app.py
- |— requirements.txt
- |— Dockerfile

1. Dockerfile.

```
FROM python:latest

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
```

2. App.py

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def hello():
```

```
return "hello world from container"
```

```
if __name__ == '__main__':  
    app.run(host="0.0.0.0", port=5000)
```

3. Run the Build command to build the image.

```
_1rv24mc078_prateek_kumar@arch ~/d/new_lab_3> sudo docker build -t prg_3 .  
[sudo] password for _1rv24mc078_prateek_kumar:  
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.  
            Install the buildx component to build images with BuildKit:  
            https://docs.docker.com/go/buildx/  
  
Sending build context to Docker daemon  3.072kB  
Step 1/7 : FROM python:latest  
latest: Pulling from library/python  
281b80c799de: Pulling fs layer  
15f14138abe4: Pulling fs layer  
378c64c44580: Pulling fs layer  
02e37abc533a: Pulling fs layer  
b1741a62ccee: Pulling fs layer  
add107facb26: Pulling fs layer  
1227bf08bd42: Pulling fs layer  
02e37abc533a: Waiting  
b1741a62ccee: Waiting  
1227bf08bd42: Waiting  
add107facb26: Waiting  
█
```

4. Output

```
(singh@singh: ~/lab4demo)  
-$ sudo docker run -p 5000:5000 program4  
* Serving Flask app 'app'  
* Debug mode: off  
WARNING: This is a development server. Do not use it in a production deployment  
* Running on all addresses (0.0.0.0)  
* Running on http://127.0.0.1:5000  
* Running on http://172.17.0.2:5000  
Press CTRL+C to quit  
72.17.0.1 - - [07/Nov/2025 04:19:53] "GET / HTTP/1.1" 200 -  
72.17.0.1 - - [07/Nov/2025 04:19:56] "GET / HTTP/1.1" 200 -  
72.17.0.1 - - [07/Nov/2025 04:23:05] "GET / HTTP/1.1" 200 -  
72.17.0.1 - - [07/Nov/2025 04:23:05] "GET /favicon.ico HTTP/1.1" 404 -  
72.17.0.1 - - [07/Nov/2025 04:26:31] "GET / HTTP/1.1" 200 -  
C
```

http://localhost:5000



A screenshot of a web browser window. The address bar shows 'http://localhost:5000' and the page content displays 'hello world'.

